
Synthetic Cultural Agents 2.0: Beyond Persona-Prompting

A Position Paper on Structural Preference Estimation for Culturally Aligned Language Models

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Status. Intermediate summary of SCA 2.0: structural conjecture, implemented DPO pipeline, Tier-1 synthetic pair recovery, and Tier-2 out-of-sample empirical evaluation on World Values Survey Wave 7 items ($N = 35$). Code & data: github.com/EconLLM-Lab/SCA2_PofW.

Abstract

Large language models are increasingly used as simulators of human judgment in economics and adjacent social sciences, yet their default alignment reflects predominantly Western, Educated, Industrialized, Rich, and Democratic (WEIRD) preference data. Earlier Synthetic Cultural Agents (SCA 1.0) addressed this gap by conditioning frontier models on ethnographic profiles at inference time. That approach is useful for piloting but is reduced-form: cultural signal lives only in the prompt, is fragile to model updates, and does not support formal identification or counterfactual analysis.

This position paper outlines SCA 2.0, a structural alternative. We treat Direct Preference Optimization (DPO) as maximum-likelihood estimation in a Bradley–Terry random-utility model, using the Global Preferences Survey (GPS) six-dimensional cultural state vector to generate culture-conditioned synthetic preference data. A student model (Llama-3.1-8B-Instruct) is fine-tuned via QLoRA with unconditioned prompts (`PROMPT_COUNTRY_CONDITIONING = False`) so that cultural dispositions are encoded into parameters rather than transient context.

We evaluate the framework across two tiers: (i) Tier-1 synthetic pair recovery ($n = 200$), where own-country adapters achieve 70.0–78.5% accuracy; and (ii) Tier-2 empirical out-of-sample evaluation on $N = 35$ unseen World Values Survey (WVS) Wave 7 items mapped to GPS dimensions. On non-WEIRD targets (Mexico), the matched MEX adapter demonstrates strong cultural specificity, outperforming the cross USA adapter on 97.1% of questions ($\Delta\text{TVD} = 0.1495$, 95% CI: $[0.0964, 0.2152]$). Conversely, on US targets, the USA adapter suffers from collision with the base model’s strong American pretraining prior. Option log-likelihood scoring exhibits systematic under-dispersion relative to human populations. We formalize the measurement argument through a Cronbach–Meehl nomological network linking the GPS source instrument, the DPO training domain, and the WVS criterion domain, and we delineate reliable applications (relative comparative statics, pre-field instrument stress-testing) from unreliable ones (absolute headcount polling, fine-tuning over already-dominant priors).

Keywords: cultural preferences; Direct Preference Optimization; Bradley–Terry; Global Preferences Survey; structural estimation; construct validity; World Values Survey

1. Introduction

Social scientists increasingly use large language models (LLMs) as low-cost simulators of survey responses, experimental subjects, and policy interlocutors. The practical appeal is clear: a model can generate thousands of responses in minutes, enabling exploratory design work before expensive fieldwork. The scientific risk is equally clear. Preference fine-tuning and human feedback data for frontier models are concentrated in WEIRD populations [8, 7]. When a model is asked to “answer as someone in Lagos, Buenos Aires, or Mexico City,” its behavior still largely reflects that default alignment—unless culture is introduced in a disciplined way.

The first generation of Synthetic Cultural Agents [4] showed that persona prompting—conditioning a frontier model on demographic and ethnographic profiles—can reproduce non-WEIRD patterns in economic experiments to a surprising degree. For piloting and hypothesis generation, that result is valuable. Methodologically, however, persona prompting is a reduced-form intervention. Cultural information lives in the context window; it vanishes when the session ends; and it provides no structural object (preferences, technology, information) that can be identified, tested, or counterfactually varied.

SCA 2.0 reframes the problem as structural preference estimation. The core conjecture is:

Cultural preferences measured by the Global Preferences Survey can be encoded into language-model policies via preference fine-tuning, so that implied utilities recover the direction of culture-conditioned choices and support formal evaluation against independent survey moments.

Three features distinguish the program from prompt engineering. First, culture is parameterized by an externally measured vector—the six GPS dimensions of Falk et al. [7]—rather than by free-form narrative alone. Second, training uses Direct Preference Optimization [11], which is mathematically a reparameterization of Bradley–Terry maximum likelihood [1, 10]; this licenses econometric language (utility, identification, overidentification) with minimal translation cost. Third, validation is hierarchical: GPS-targeted synthetic pairs identify the cultural state; World Values Survey (WVS) items are reserved for out-of-sample evaluation; held-out behavioral tasks assess generalization.

This document serves as both a position paper and a report on completed out-of-sample empirical evaluations on WVS Wave 7 data. Its goals are to (i) state the conjecture and design cleanly, (ii) present an honest reading of Tier-1 and Tier-2 empirical evaluations, (iii) frame construct validity for latent cultural variables in LLMs, and (iv) demarcate reliable applications from invalid downstream uses.

2. Related Work

Three short literatures motivate the design.

The GPS [7] supplies incentivized measures of patience, risk taking, positive and negative reciprocity, altruism, and trust across 76 countries—a low-dimensional, cross-culturally comparable parameter vector. Country means are not individuals (within-country variance is often large), but they are a defensible starting point for population-level simulation. A growing experimental literature uses LLMs as simulated respondents [9, 5]; much of it is reduced-form role prompting, the family to which SCA 1.0 belongs. Complementary work documents cultural gaps in model outputs [2, 6] but typically stops at evaluation rather than estimation. DPO [11] optimizes a policy so preferred responses receive higher probability than dispreferred ones relative to a reference model; under standard assumptions this coincides with Bradley–Terry likelihood for a scaled log-probability score—the bridge to structural econometrics.

What is missing—and what SCA 2.0 targets—is a pipeline that grounds preference data in an external cultural instrument, fine-tunes open-weight models rather than relying on prompts alone, and subjects the resulting agents to rigorous out-of-sample evaluation.

3. Methodology

3.1 Primitives

Let x denote a decision scenario (prompt) and y a free-text response. A cultural state is a vector

$$z = (\delta, \gamma, \xi^+, \xi^-, \alpha, \tau)^\top \in \mathbb{R}^6, \quad (1)$$

corresponding to time preference (patience), risk taking, positive reciprocity, negative reciprocity, altruism, and trust from the GPS. An SCA is a policy $\pi(\cdot | x; z)$ whose behavior is governed by a separable reward

$$R(x, y; z) = r_{\text{task}}(x, y) + \phi(z)^\top m(x, y), \quad (2)$$

where r_{task} captures generic competence (coherence, relevance), $m(x, y) \in \mathbb{R}^6$ is a vector of measurable behavioral features of the response, and $\phi : \mathbb{R}^6 \rightarrow \mathbb{R}^6$ maps GPS scores into reward weights.

The structural assumption. Equation (2) is a *structural* (not descriptive) assumption: it claims that the country-specific component of an agent’s behavior is a linear function of six externally measured preference traits, mediated by observable features of the response. Two testable implications follow. First, *separability*: the task term r_{task} is country-invariant, so any cross-country behavioral difference must load on $\phi(z)^\top m(x, y)$. Second, *dimensional structure*: only six dimensions of cultural variation are permitted to matter. Both implications are falsifiable—the first by the matched-versus-cross specificity design (Section 4), the second by out-of-sample items that “should not” be explained by any dimension and by items whose loading is measured rather than imposed.

3.2 Choice model

We treat the agent’s response as a discrete choice over a finite option set $\mathcal{Y} = \{y_1, \dots, y_K\}$ (the response codes of a survey item or the two completions of a preference pair). A Luce–Bradley–Terry–McFadden random-utility model sets

$$\Pr(y_k | x; z) = \frac{\exp(R(x, y_k; z))}{\sum_{j=1}^K \exp(R(x, y_j; z))}, \quad (3)$$

which is the softmax over option rewards. For a pair $\{y^w, y^\ell\}$, equation (3) collapses to the familiar logistic paired-comparison probability

$$\Pr(y^w \succ y^\ell | x) = \sigma(R(x, y^w; z) - R(x, y^\ell; z)), \quad (4)$$

with $\sigma(u) = (1 + e^{-u})^{-1}$.

Equation (3) is the exact object the evaluation pipeline estimates: the option-likelihood protocol (Section 4) computes $\hat{p}_k \propto \exp(\log\text{prob}(y_k | x))$ over all valid response codes and compares the resulting distribution to the empirical population distribution. The LLM is therefore used as a *choice-probability simulator*, not a text generator—a deliberate design choice that makes the model’s output directly comparable to the random-utility object in equation (3).

3.3 DPO as structural estimation

Given preference data $\{(x_i, y_i^w, y_i^\ell)\}_{i=1}^n$ in which y^w is preferred to y^ℓ under cultural state z , the Bradley–Terry likelihood is

$$\mathcal{L}(R) = \prod_{i=1}^n \sigma(R(x_i, y_i^w; z) - R(x_i, y_i^\ell; z)). \quad (5)$$

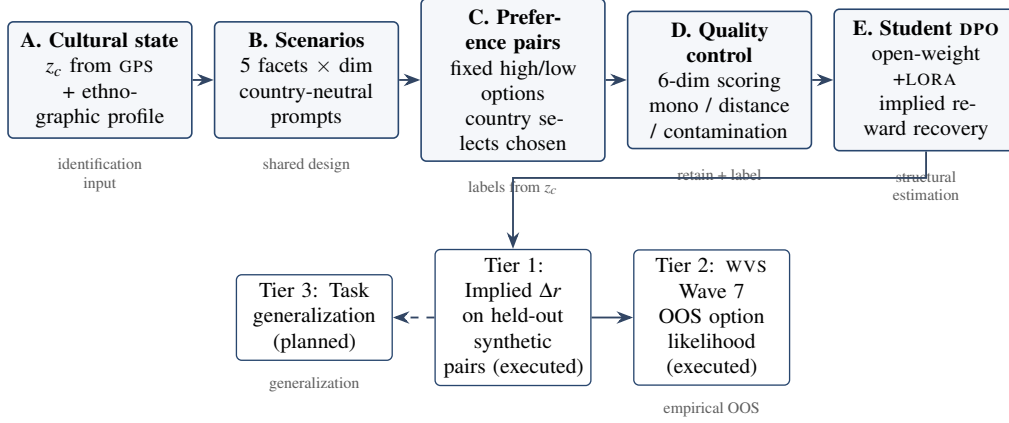


Figure 1: SCA 2.0 pipeline architecture. Solid lines mark executed pipeline stages. Culture enters as an external GPS vector z_c ; synthetic pairs are generated, quality-controlled, and fine-tuned via DPO; model validation proceeds through Tier-1 synthetic pair recovery and Tier-2 empirical WVS Wave 7 option-likelihood evaluation.

Direct Preference Optimization [11] maximizes this likelihood while reparameterizing the reward difference through the log-ratio of the trained policy to a frozen reference policy:

$$R(x, y_1; z) - R(x, y_2; z) = \beta \left[\log \frac{\pi_\theta(y_1 | x)}{\pi_{\text{ref}}(y_1 | x)} - \log \frac{\pi_\theta(y_2 | x)}{\pi_{\text{ref}}(y_2 | x)} \right], \quad (6)$$

where π_{ref} is a frozen reference model and $\beta > 0$ is a scale temperature. Equation (6) is the empirical analogue of the structural reward in equation (2): after training, the analyst can compute, for any held-out pair, the *implied reward margin* $\widehat{\Delta r} = \beta [\log\text{-ratio of chosen} - \log\text{-ratio of rejected}]$, which is the estimated utility difference under the country-specific z .

Identification and its limits. Equation (6) identifies reward *differences* up to scale (β) and location (any additive constant cancels in paired comparisons). The two structural features we *can* recover are (i) the sign of Δr —which option a country’s policy ranks higher—and (ii) the own-versus-other contrast, i.e., whether an adapter trained on country c ’s labels assigns systematically larger margins to c ’s chosen options than an adapter trained on country c' . We cannot recover the absolute level of R nor, in general, the individual coordinates of $\phi(z)$ without additional moment conditions—a limitation we make explicit rather than obscure.

3.4 Pipeline Architecture

Figure 1 summarizes the pipeline architecture across data generation, fine-tuning, and out-of-sample validation.

3.5 Construct validity and the nomological network

Because z is a *latent* construct—nobody observes “patience in Mexico” directly—the scientific status of the pipeline hinges on construct validity. We adopt the classical formulation of Cronbach and Meehl [3]: a construct is validated by embedding it in a *nomological network*, a set of theoretical relations linking the construct to (a) other constructs and (b) observable measures. Validation is then a matter of testing whether the observed correlations behave as the network predicts.

The network. For the latent construct “country-level economic preference disposition,” the network has three nodes—three families of observables that the construct must systematically relate to:

1. **Source domain (identification):** the GPS instrument z_c (Falk et al., 2018), an incentivized, cross-nationally comparable measure of the six preference dimensions. This is the *defining* measure of the construct: z_c is what we mean by “Mexico’s preference profile” in this paper.
2. **Training domain (internal manifestation):** the synthetic preference pairs $\{(x_i, y_i^w, y_i^\ell)\}$ whose labels are generated by z_c , and the implied reward margins $\widehat{\Delta r}$ recovered from the trained policy.
3. **Criterion domain (external manifestation):** the World Values Survey (WVS) item-response distributions P_{WVS} , an independent, human-administered survey that was *not* used in training and whose items are mapped to the six dimensions via a curated schema (`question_map_wvs_edited.csv`).

The network postulates three links, each with a distinct falsifiable prediction:

1. **Link 1 (source $\rightarrow$ training):** preferences implied by z_c must rank y^w above y^ℓ on held-out pairs. Falsified if the trained policy fails to recover its own labels (Tier-1 preference accuracy ≤ 0.5).
2. **Link 2 (training $\rightarrow$ criterion, convergent validity):** an adapter trained on z_c must reduce distributional error against P_{WVS} for country c on unseen items, relative to the base model—because both the synthetic labels and the WVS responses are driven by the same underlying z_c . Falsified if fine-tuning does not improve (or degrades) empirical fit.
3. **Link 3 (criterion-domain discrimination, discriminant validity):** an adapter trained on z_c must fit c ’s WVS distributions better than an adapter trained on $z_{c'}$, on the *same* items, with country names withheld from the prompt. Falsified if the matched adapter does not dominate the cross adapter (matched-vs-cross specificity ≤ 0.5).

How the network handles negative evidence. The network is a *joint* claim, and its links can fail independently—which is precisely what the data show. The USA adapter *passes* Link 1 (Tier-1 accuracy 0.70) but *fails* Links 2–3 on US WVS items: the base model already encodes a strong US pretraining prior, so the adapter’s additional synthetic signal over-corrects rather than refines. This is not evidence against the construct; it is evidence that the *source domain* is redundant with the model’s pretraining for WEIRD populations. The same network then correctly predicts that the MEX adapter—trained on a population outside the pretraining prior—should pass all three links, which it does (Tier-1 accuracy 0.785; matched-vs-cross specificity 97.1% on MEX items; dimension-level convergence such as Altruism TVD falling from 0.450 to 0.152).

Reliability. Reliability—consistency of measurement—is assessed within the same framework: bootstrap confidence intervals over questions (2,000 replications) for every reported metric; calibration tables (predicted versus empirical option frequencies, ECE); and signed moment diagnostics (mean and dispersion bias) that reveal the structural under-dispersion of softmax option scoring. These diagnostics do not merely qualify the results; they are part of the nomological network, because they bound how much of the observed fit is construct signal versus scoring artifact.

4. Empirical Results

4.1 Tier-1: Synthetic Preference Recovery

Table 1 reports Tier-1 results on held-out synthetic preference pairs ($n = 200$ per adapter–country cell). Both the Mexico and US adapters recover their own culture-conditioned synthetic labels well above chance (78.5% and 70.0% preference accuracy, respectively), while cross-country accuracy remains near chance (52.0% and 45.5%).

Table 1: Tier-1 Cross-evaluation of country adapters on held-out synthetic preference pairs ($n = 200$ per cell). Preference accuracy is the share of pairs with positive implied reward margin ($\Delta r > 0$).

Adapter	Eval Data	n	Mean Δr	Pref. Accuracy	Mean $P(\text{chosen})$
Mexico Adapter	Mexico	200	+1.70	0.785	0.714
Mexico Adapter	United States	200	−0.22	0.520	0.501
USA Adapter	Mexico	200	−0.20	0.455	0.464
USA Adapter	United States	200	+0.78	0.700	0.632

4.2 Tier-2: World Values Survey Out-of-Sample Evaluation

Tier-2 tests frozen adapters on $N = 35$ unseen Wave 7 WVS survey questions under unconditioned prompts (PROMPT_COUNTRY_CONDITIONING = False). Generated option distributions $\hat{P}$ are evaluated against survey-weighted human response distributions P_{WVS} . Table 2 summarizes the primary evaluation metrics.

Table 2: Tier-2 Out-of-Sample Evaluation on $N = 35$ unseen WVS Wave 7 survey questions under unconditioned prompts. Specificity represents the fraction of questions where the matched adapter achieves lower error than the cross adapter.

Target Population	Model / Adapter	TVD ↓	JSD ↓	ECE ↓	Specificity (% Matched > Cross)
Mexico Data (MEX WVS)	Base Model (Llama-3.1-8B)	0.5211	0.2165	0.0919	—
Mexico Data (MEX WVS)	MEX Adapter (Matched)	0.4882	0.2010	0.0967	97.1% (34/35)
Mexico Data (MEX WVS)	USA Adapter (Cross)	0.6377	0.2965	0.1220	—
USA Data (USA WVS)	Base Model (Llama-3.1-8B)	0.3958	0.1439	0.0654	—
USA Data (USA WVS)	MEX Adapter (Cross)	0.3888	0.1405	0.0719	—
USA Data (USA WVS)	USA Adapter (Matched)	0.5258	0.2246	0.0958	11.4% (4/35)

Key Finding 1: Cultural Specificity on Non-WEIRD Targets. On Mexico WVS data, the matched MEX adapter achieves striking cultural specificity, outperforming the cross USA adapter on 97.1% of questions (34/35 items). The mean difference in Total Variation Distance ($\text{TVD}_{\text{cross}} - \text{TVD}_{\text{matched}}$) is +0.1495 (95% CI: [0.0964, 0.2152]). Because prompts omit country names, this improvement proves that country-specific dispositional priors were internalized into adapter weights.

Key Finding 2: Pretraining Prior Collision on USA Targets. On USA WVS data, the matched USA adapter performs *worse* than both the base model ($\text{TVD} = 0.5258$ vs. 0.3958) and the cross MEX adapter ($\text{TVD} = 0.3888$). Only 11.4% of questions favored the matched USA adapter. Base Llama-3.1-8B-Instruct already carries a massive American pretraining prior; fine-tuning on synthetic US pairs causes over-correction and distributional drift rather than refinement.

Key Finding 3: Structural Under-Dispersion. Across all models, option log-likelihood scoring suffers from severe **under-dispersion** (dispersion bias ranging from −1.15 to −1.63). Softmax normalization concentrates probability mass on modal choices, underestimating real-world human population variance.

5. Scope of Methodology & Reliable Applications

An honest reading of the methodology requires defining where the current pipeline is scientifically reliable versus where downstream deployment would be invalid.

5.1 Reliable Applications

1. **Implicit Preference Internalization:** Fine-tuning via unconditioned DPO successfully embeds non-WEIRD behavioral traits directly into model parameters, eliminating reliance on brittle context-window persona prompts.
2. **Relative Comparative Statics:** Valid for evaluating ordinal shifts across non-WEIRD populations (e.g., tracking how policy choices shift when Altruism TVD improves from 0.450 $\rightarrow$ 0.152).
3. **Pre-Fielding Instrument Stress-Testing:** Useful for probing survey questionnaires for cultural sensitivity and bias before launching expensive human fieldwork.
4. **Zero-Shot Option Probability Probing:** Provides a clean diagnostic harness for measuring model likelihood assignments without generation fluff.

5.2 Unreliable Applications & Boundaries

1. **Absolute Headcount Forecasting & Polling:** Cannot be used to predict absolute percentage shares in election or survey polling due to structural softmax under-dispersion.
2. **Fine-Tuning Over Already-Dominant Priors:** Fine-tuning adapters for populations already heavily represented in pretraining (e.g., US/WEIRD populations) causes prior collision and performance degradation.
3. **Free-Form Generative Personas:** Option log-likelihood validation does not guarantee that open-ended chat generations will sound authentically human.

6. Conclusion and Next Steps

SCA 2.0 demonstrates that cultural alignment can be treated as structural preference estimation. While the methodology successfully internalizes non-WEIRD preference distributions (Mexico), it surfaces key boundaries regarding pretraining prior collision (USA) and softmax under-dispersion. Immediate engineering next steps include implementing temperature calibration to correct dispersion bias, expanding item coverage across AmericasBarometer, and regularizing DPO training against base pretraining priors.

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